System Administration and Maintenance

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IP CLASS

IP Class	Address Range	Maximum number of networks
Class A	1-126	126 (2^7-2)
Class B	128-191	16384
Class C	192-223	2097152
Class D	224-239	Reserve for multitasking
Class E	240-254	Reserved for Research and development

IP CLASS

Special IP Addresses

There are also some special-purpose IP addresses that don't follow the usual structure:

- **Loopback Address**:
  - The loopback address **127.0.0.1** is used to test network connectivity within the same device (i.e., sending data to yourself).
  - Often called "**localhost**."

- **Broadcast Address**:
  - The broadcast address allows data to be sent to all devices in a network.
  - For a typical network with the IP range **192.168.1.0/24**, the broadcast address would be **192.168.1.255**.

- **Multicast Address**:
  - Used to send data to a group of devices (**Multicast**).
  - For example, **233.0.0.1** is a multicast address.

IP addresses are also classified into different classes based on their range and intended use:

- **Class A (1.0.0.0 - 127.255.255.255)**:
  - Used for very large networks (like multinational companies).
  - Supports up to 16 million hosts per network.
  - **Example:** 10.0.0.1 (Private IP in this class).

- **Class B (128.0.0.0 - 191.255.255.255):**
  - Used for medium-sized networks, such as large organizations.
  - Supports up to **65,000 hosts** per network.
  - **Example:** 172.16.0.1 (Private IP in this class).

- **Class C (192.0.0.0 - 223.255.255.255)**:
  - Used for smaller networks, like small businesses or home networks.
  - Supports up to 254 hosts per network.
  - **Example:** 192.168.1.1 (Private IP in this class).

- **Class D (224.0.0.0 - 239.255.255.255)**:
  - Reserved for multicast groups (used to send data to multiple devices at once).
  - Not used for traditional devices or networks.

- **Class E (240.0.0.0 - 255.255.255.255)**:
  - Reserved for experimental purposes and future use.

NAT (Network Address Translation)



Static IP & Dynamic IP

- **Static IP Addresses:**
  - These are permanently assigned to a device.
  - Typically important for servers or devices that need a constant address.
  - Reliable for network services that require regular access.
  - Such as websites, remote management.

- **Dynamic IP Addresses:**
  - Temporarily assigned from a pool of available addresses by the **Dynamic Host Configuration Protocol (DHCP).**
  - Cost-effective and efficient for providers, perfect for consumer devices that do not require permanent addresses.

THANK YOU